Phase 4 - Econometric Modeling

Phase 4 fits a regression model that decomposes per-course opportunity cost into its structural and geographic determinants. The model is a logarithmic OLS regression with heteroskedasticity-robust (HC1) standard errors, fit independently on each of the 100 imputed datasets produced by Phase 3 and then pooled across imputations using Rubin's Rules. The result is a set of three statistically valid coefficient estimates per language (Python, R, Julia), with confidence intervals that correctly account for both within-imputation sampling variance and between-imputation imputation variance.

I deliberately specified a parsimonious model. The dependent variable is the natural log of per-course opportunity cost (`log(osm_acreage × Baseline_Value_Per_Acre)`), and the right-hand side carries two regressors: the number of holes (a continuous structural proxy for course size) and a binary indicator for whether the course sits in an urban county (RUCC 1–3 versus 4–9). I considered adding more covariates and rejected the addition: course type (public/private/municipal) and ownership type would have absorbed variance that the urban indicator already explains structurally, and including latitude/longitude would have produced spatial autocorrelation issues without methodologically clean correction. The specification I chose produces clean coefficient interpretations and tight cross-language convergence.

For each language, the pipeline runs in two steps: (1) fit OLS with HC1 robust standard errors on each of the 100 imputed datasets and serialize the coefficient and variance vectors; (2) pool across the 100 fits using Rubin's Rules to compute pooled point estimates, total variance ($V_T = V_W + (1 + 1/M) \cdot V_B$), pooled standard errors, Fraction of Missing Information (FMI), and Barnard-Rubin adjusted degrees of freedom. Each fit drops 34 rows with missing covariate data, leaving N = 16,258 observations per imputed dataset.

Key result

The pooled regression produces three coefficient estimates that are extraordinarily consistent across the three languages. The Holes coefficient is approximately $\beta_1 \approx 0.05$ in all three (0.047 Python, 0.053 R, 0.048 Julia), implying that each additional hole increases log opportunity cost by about 5%. The urban indicator coefficient is approximately $\beta_2 \approx 4.0\text{--}4.2$ in all three (4.172 Python, 4.001 R, 4.158 Julia), implying that an urban course's opportunity cost is roughly $\exp(4.1) \approx 60$ times that of a comparable rural course on a per-acre basis. All coefficients are statistically significant at $p < 0.001$ in every language. The model R^2 ranges from 0.70 (R) to 0.77 (Python), with the variation attributable to differences in the underlying MICE imputation backends across the three stacks.

The 60× urban-rural ratio is partially mechanical: the hybrid valuation algorithm in Phase 1 assigns urban courses the FHFA residential price and rural courses the USDA agricultural price, and the FHFA-USDA per-acre ratio is itself approximately 60× in many counties. The regression is therefore best read as a *decomposition* of the bifurcated baseline rather than a causal estimate of how a course's value would change if it were relocated across the rural-urban boundary. I treat this interpretive caveat as important and carry it forward into the thesis prose in Section 5.3.

What was solved

The Julia HC1 implementation required hand-rolling the sandwich estimator because `GLM.jl` did not expose a direct robust-covariance API at the version pinned for this project. My fix computes `(n/(n-k)) · (X'X)⁻¹ · X' · diag(ê²) · X · (X'X)⁻¹` directly from the model matrix and residuals returned by `GLM.lm`, matching the HC1 formula used by `statsmodels` in Python and `sandwich::vcovHC` in R. The resulting Julia standard errors agree with R and Python to within rounding error across all three coefficients.

Outputs flow forward into Phase 5 (Hawaii micro-study), which uses Phase 4's per-course pooled estimates to compare model HBU values against parcel-level tax assessments, and into Phase 6, which generates the forest plot, marginal-effects chart, and other visualizations from the pooled regression results.